

SE Lab 3

Component Modelling & Architectural Pattern Selection

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Architecture Selection

We chose Layered Architecture for the Peer Skill Exchange & Mentorship Network.

Architectural Choice

Layered Architecture, organised into a Presentation Layer (User Interface Component), a Business Layer (Session Manager, Time-Credit Service, Rating & Trust, and Calendar Sync Components), and a Data Layer (Database Component).

Two Reasons for This Choice

1. Natural three-tier separation. The platform splits cleanly into what learners and mentors see (browsing skills, requesting sessions), the rules that govern exchanges (booking, credit transfer, ratings), and persistent storage of users, sessions, and credit balances. Each of these maps directly onto one layer, so no artificial splitting is needed.

2. Consistent time-credit logic. Time-credit transfer is the core trust mechanism of a time-banking platform. Keeping credit checks, session confirmation, and rating validation together in one Business Layer makes it straightforward to guarantee that credits move only after both parties have confirmed and rated a session. Distributing this logic across independent services would make that guarantee much harder to enforce.

Security Advantage

The Presentation Layer never accesses the Database Component directly. Every change to a credit balance or rating passes through the Business Layer, where it is validated first. This closes off any path for a compromised or malicious client to alter credit balances or ratings directly. The Business Layer also acts as a single enforcement point for checks such as insufficient-credit rejection.

Performance Benefit

Booking a session involves several steps: checking credit balance, holding the time slot, and syncing with the calendar. Because these components all sit within one Business Layer, they exchange calls in-process and share request context, avoiding the network round-trip latency that a microservices split would add at each step. This keeps booking responsive, in line with the calendar-sync performance target set in NFR-001.